CSC 312 - Software Engineering Group Project 2026

This group project prioritizes **the application of foundational software engineering aspects** over algorithmic or computational complexity. You will gain hands-on experience navigating the software development lifecycle (SDLC/SE process), **utilizing industry-standard SE processes, principles, methods, umbrella activities and tools in producing a quality (i.e., ISO 205010 compliant) product/service**. A primary objective of the group project is to develop essential team dynamics and professional collaboration skills. Consequently, your assignment will heavily emphasize the rigor of your development process, clear adherence to SE aspects and the quality of your project documentation rather than the intricacy of the underlying problem.

Please complete the following project in the groups you have formed.

Your group is free to propose and choose to develop any system/application/app, in any domain (area) for your SE Group Project - so it is a project of your choosing. You are welcome to include aspects from the other sections of CSC311/312 (e.g. AI, DB and ML) within this project. Make any reasonable assumptions you feel necessary to make the system/application/app work. Include any other features/functions you may deem necessary to complete/improve the application (i.e. you can expand the project scope if you need to as can be done in an Agile process).

The final project needs to be completed/uploaded by **Sunday 8 November 2026**. Your work products will be uploaded via iKamva assignments. **NO ZIP or DOC FILES will be accepted.**

The Group Project will be graded/marked continuously as set out below.
The project will involve modelling and designing (e.g. Use-cases, Class (or function) diagrams, pseudocode, etc.) and then implementation in Java or any other implementation language of your choice (e.g. Python). Choose any development environment such as NetBeans IDE, Eclipse or some other equivalent development environment (e.g. for a mobile app development) which supports UML. You can do the project on your laptop/PC/mobile device - start working in your groups as soon as possible. The GLA/tutors will be available for consultation and help – see their contact details in the course description on iKamva.

Select and apply one of the SE Process models ("road maps") so that you do the project in a disciplined way - see Chapter 2/3/4 slides. For example, you may choose to follow the Waterfall, Incremental, Spiral, Agile (e.g. Scrum) or some other process model of your choosing that is appropriate. Choose a project manager, group leader or scrum master to provide leadership to the group.

Produce a suitable document/report (in PDF) which captures the process and other fundamentals of the software engineering process. This report format will use the ACM conference layout guideline (e.g. two columns per page, font size 9 for general text, etc.) – this template guideline and a lesson on using it has been supplied on our iKamva resources (item 11 under SE). **Here below is a report guideline to help structure your document and content** based on the generic process model – include only what's applicable for your project:

> **Communication** – scope (objectives, requirements, stakeholders, assumptions, constraints, functional, non-functional, etc.).
> **Planning** – process model selected; work and technical tasks to be done; explain why you chose a particular process model and an honest reflection on the good and bad experiences of the process you had within your particular group. Where applicable, include comments on the principles of SE, methods, SE goals (quality) and umbrella activities. (You need not comprehensively have all these elements in your project but at least some relevant ones.)
> **Modelling** – **analysis** (more detailed Use-Cases, data, functions) and **design** (class diagrams, activity diagrams, pseudocode, etc.); UID
> **Construction** – code description and testing information (e.g. test cases)
> **Deployment** – delivery/installation information

Apply at least two of the principles, methods, goals, umbrella activities and quality attributes applicable for software engineering - particularly those that are relevant for your particular project. UML must be used where appropriate (e.g. Use case diagrams, class diagrams, etc.). Where applicable, you may of course model the data in an ERD.

When done we will assess your **report (PDF)** containing your software development process followed

(and reflection, Use case diagram, class (or function, other) diagrams), code (in PDF) and a **presentation and demonstration video (MP4)** of the application. This mark is an important component of your SE semester mark component. Students who did not participate (as reported by the group leader) in the SE group project, will receive zero for the SE group project mark (40% of SE mark).

There is lots of online help available (e.g. YouTube) for UML features, NetBeans, Eclipse, Java, etc.

For those who are implementing the application in a non-OOD/OOP manner - you will instead have functional diagrams (i.e. a functional decomposition) rather than class diagrams to ensure that your model matches your implementation. (All designs will of course have Use-cases initially.)

Please email me if there are any uncertainties.

Best wishes.
Dr MJ Norman

Steps for the SE Group Project:

Step 1: Group leader only must upload the group details to an iKamva assignment called "SE Groups for 2026" by Sunday 30 August 2026 - with the **group members** (name and surname and student number) and the **group name**.

Step 2: Group leader to upload the Project Description by Sunday 6 September 2026

Software Engineering Project Proposal Assignment

Overview

This assignment requires your team to propose a realistic software project, define the software process model you will use to build it, and outline its core technical and functional scope.

Assignment Breakdown & Requirements

Your proposal must contain the following four sections – use these as headings (and subheadings) to structure your proposal:

1. Project Abstract & Problem Statement (Max 100 words for this item)

Problem Definition: Clear explanation of the specific problem or real-world issue your project addresses.

Core Aim: A brief summary of what the software aims to achieve and its proposed solution.

2. Stakeholder Identification

Identify the key primary and secondary stakeholders (e.g., end-users, administrators, system integrators) and briefly state how the system serves their needs.

3. Software Process Model & Software Engineering Practices

Process Model: Select and justify one recognized process model (e.g., Agile/Scrum, Waterfall, Spiral, V-Model, etc).

Process Elements: For each of the following categories, select and describe at least two relevant elements you will apply:

SE Principles
SE Methods/Techniques
SE Goals (from ISO 25010)
SE Umbrella Activities

Toolchain: Specify the tools you will use to support your SE process.

4. Project Scope & Requirements

Ensure the project scope is realistic for the duration of this course (Term 3 and Term 4) in terms of the following two aspects:

Functional Requirements: Feature list, user tasks, and core functionalities.

Non-Functional Requirements: Key system constraints (e.g., performance, security, usability, scalability – essentially from the SE Goals above).

Step 3: Start working on the SE Group Project as outlined above using a SE Process incorporating UML.
Also apply the principles of SE, methods, SE goals (quality) and umbrella activities. (You need not comprehensively have all these elements in your project but at least some relevant ones from each of these aspects as set out in Step 2 assignment.)

Step 4: User Model (Requirements and Analysis) – document uploaded by Sunday 27 September 2026 [50]

All documents/reports in ACM conference format, e.g. two columns per page. See folder *11. Report format template and* lesson for template and lesson on using it. Documents/reports uploaded will be checked by Turnitin (includes AI-detection).

Identify the users' needs for this project. You could do this in a number of ways. For example, suppose you were to design a booking system - in this case, you could observe people using ticket agents, think about your own experience of purchasing tickets, look at existing websites/apps for booking tickets, talk to friends and family about their experiences, and so on. Conduct similar activities with respect to your project. Record your data carefully.
See Chapter 15 User Interface Design or the slides (in SE slides) or the chapter summary (in course outline) which will guide all aspects of the UID.

- **Tasks**
  - **User model** — a profile of all end users of the system and the tasks they should be able to perform; you can model these in a UML Use-case diagram.
  - **Mental model** (system perception or conceptual model) — the user's mental image of what the interface is; e.g. if you are designing an online shop (e.g. Takealot.com; Checkers Sixty60) you use the concept or mental picture of a physical shop with a trolley (cart) and isles and departments or categories (e.g. bakery) which users can relate to since they are familiar with physical (brick 'n mortar) shops where they have purchased items
  - **Requirements Analysis** – identify and/or model the data, function, behaviour. Your UML Use case may be used and/or Class (or function) diagrammes – these class diagrams will be at high (abstract) level – more details will be added in the next assignment (Design).

Using the data gathered above and your subsequent interpretation and analysis, identify different kinds of requirements for the project.
A use case is a diagram or written description of how users will perform tasks on your website/app. It outlines, from a user's point of view, a system's behavior as it responds to a request. Each use case is represented as a sequence of simple steps, beginning with a user's goal and ending when that goal is fulfilled.

Document and write up your project under the following suggested headings as they emerge during the project progress - User Model, Mental Model, Requirements Analysis and in later steps you will document the Design Model, Implementation Model. You can include other headings as you deem appropriate (e.g. Introduction, scope, etc.).

Continue with the application of the processes, principles, methods, goals, umbrella activities and tools associated with software engineering.

Step 5: Design Model plus a Prototype – document (PDF) plus prototype demo (3-min max in MP4) uploaded by Sunday 11 October 2026 [50]

Design Model - See Chapter 12 (Design Concepts) – data design (e.g. ERD if applicable), class design, architectural design, interface design and component level (e.g. pseudocode) design as well as Chapter 15 (specific focus on User Interface Design). User Interface Design (UID) Process – see the lecture slides on Chapter 15 (User Interface Design) about Interface Design Models.

Develop a Prototype - Using a mockup tool or a software-based prototyping tool like Figma/Adobe XD/Justinmind/etc or even a web authoring tool (e.g. WordPress), develop a prototype which demonstrates the key functionality of your application/app - this type of prototype is a throw-away prototype and will then be followed by actual coding.
OR
You can create the prototype using actual program code/scripts showing the key functionality - this type of prototype will then evolve into the completed application/app.

Continue with the application of the **SE aspects** - process, principles, methods, goals, umbrella activities and tools associated with software engineering. Write in your document about these.

Steps 6 and 7: Construction (Implementation/Coding and Testing) – project document (PDF), demo of application/app (MP4 – 3min max) and project presentation (MP4 – 5min max) uploaded by Sunday 8 November 2026

Only PDF's and MP4's as requested will be accepted – no ZIP files or other formats will be graded.

Implement the design using software tool(s) of your choice (e.g. an integrated development environment (IDE) – e.g. NetBeans, Microsoft Visual Studio, Adobe Flex Builder, Eclipse, Android IDE, Xcode, etc). Write in your document about the hardware, software tools used.

Apply at least one testing strategy (appropriate for your application/app) to ensure you have satisfied the user requirements and to uncover any errors/bugs. Testing is the process of exercising a program/application/app with the specific intent of finding errors (e.g. in functional requirements; in non-functional requirements; and an indication of quality (e.g. usability)) prior to delivery to the end user. There are many testing strategies available to choose from – see Chapter 22 – only select one testing strategy. Add a new section to your report (PDF) entitled "Testing" documenting this final step.

Continue with the application of the **SE aspects** - process, principles, methods, goals, umbrella activities and tools associated with software engineering. **Write/update in your document about these explaining why you chose a particular process model and give an honest reflection on the good and bad experiences of the process you had within your particular group**. Add a heading "Reflection on the SE Process" in your document report.

About the project presentation:

- A **5-minute slide presentation/narration about your project in MP4 format.**
  The 5-minute Presentation should start with some background information – it should include the following – one slide on each of the following should do:
  - the project title, group name and who the group members are
  - tell us about the specific application/app that was chosen
  - what the requirements were and the resulting requirements analysis
  - what the design components were
  - what the implementation details for your system/application/app
  - what type of testing was done
  - what SE aspects were applied

There is no need to upload an executable file/package – your demo in Step 6/7 (Construction) covers this.

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